Project Proposal

Problem Statement

The bandwidth of Bluetooth Low Energy (BLE) 6.0 is too low to support transmission of raw neural extracellular electrophysiology data (hereby referred to as continuous neural data) sampled at the industry standard from more than two recording channels at once for offline spike sorting and consequent neuroscientific analysis [insert reference].

Need Derived from Problem Statement

Compress continuous neural data on the head stage of a rodent without domain specific loss of fidelity for wireless transmission.

Design Requirements and Technical Specifications

Description

The proposed system consists of two overarching parts, or phases, under different constraints: one offline phase and one online phase. The offline phase runs once on a workstation where it sorts a reference recording and trains the classifier. The online phase, running on the SoC, performs inference against the resulting pre-trained model. Only the online phase is subject to the power, latency, and memory constraints that motivate this thesis. The requirements described below apply exclusively to the online phase unless otherwise noted.

The online phase realizes the declared need through five procedural operations: it *acquires* extracellular voltage from multiple recording channels, *detects* spike events within the acquired signal, *records* the time at which each detected spike occurs, *classifies* each spike by assigning it a unit identity, and *transmits* each unit identity, paired with its timestamp, over a wireless link. These five operations constitute the functional requirements of the online system. However, despite realizing the declared need,the operations making up the functional requirements of the online phase are dependent on the offline phase.

Paragraph about the procedural operations of the offline phase, which we need functional requirements for as well I guess.

Each functional requirementimposes one or more technical requirements, grouped by the operation that produces them which is presented in Table 1.

Technical requirements were first derived independently for each stage of this online pipeline. One requirement, however, resisted this per-stage decomposition: that no spike be lost. A sample or detected spike dropped anywhere in the chain — whether during acquisition, transmission to the classifier, or over the wireless link — compromises the final result identically. This requirement is therefore treated as a system-level requirement spanning the full pipeline, summarized separately in Table 2, rather than attached to any single functional requirement in Table 1.

Table 1: Per-stage technical requirements of the online pipeline.

Label	Requirement	Specification	Quantitative value	Measurement method
T1	≥ 3 channels amplified and digitized	channel count	≥ 3; target 32	count channels in acquired stream
T2	Sample fast enough for the spike band	per-channel sample rate	≥ 12.5 kS/s; design 25 kS/s	FFT of a known replayed signal
T3	Digitization keeps up across channels	aggregate conversion rate	$\geq$ channels $\times$ per-channel rate	sustained conversion count
T4	Resolve spikes above the noise floor	resolution / SNR	16-bit; SNR [TBD]	noise and SNR on a known input
T5	Link carries the digitized stream	ADC$\rightarrow$MCU throughput	$\geq$ channels $\times$ rate $\times$ bits	dropped-sample count
T6	Spikes separated from the LFP	high-pass corner	$\sim$300 Hz	filter frequency response
T7	Detect spikes reliably	sensitivity / false-positive rate	TPR [TBD]; FPR [TBD]	compare to Kilosort 4.0
T8	A timestamp is recorded per spike	timestamp presence	one per detected spike	inspect output stream
T9	Each spike assigned a unit identity	unit ID from library	one of K units	inspect output
T10	Classify better than chance	accuracy vs. Kilosort	$> 1/K$ (chance); [TBD]	confusion matrix vs. Kilosort
T11	Packet carries ID and timestamp	packet content	5 B (1 B ID + 4 B ts)	inspect received packets